

Fast-Moving Drone For Public Announcements

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Racing drones are fast and can therefore cover long distances quickly. We can use racing drones for other applications as well such as public announcement as described in this article. The KK.2.2 flight controller used in 'Make Your Own Quadcopter Drone' DIY project in January issue had some problem with setup and stabilisation of the drone. But we can make a smart public announcement drone using QAV250 frame and CC3D OpenPilot hardware. The author's prototype is shown in Fig. 1.

QAV250 drone frame being small in size, it requires a high RPM and kV rated motor for flying at a high speed

and flexibility in movement. We need a smaller size flight controller and propeller that can fit in the drone frame. A motor having higher than 2,200kV rating and propellers between 12.7 and 15.2cm (5 to 6 inch) in size would do the job. But when 'A2212' 2200kV motors were used with 12.7cm propellers, it did not produce enough thrust to fly fast enough. Therefore larger '6045' propellers of 15.2cm size with 10.2cm pitch (6-inch with 4-inch pitch) were used.

Assembly

First of all, assemble the frame of the drone. The QAV250 drone frame has two base plates, four arms for the motors, one top plate, and four

landing gears. Place all the four arms between the two base plates and screw them. Add the landing gears on the four arms and then fix the top plate on the base plate. The drone frame is now ready to equip with electronic controller and other components on the base plate. The assembled frame is shown in Fig. 2.

Connect ESC wires to the power distribution board along with the XT60 plug, as shown in Fig. 3. Place the power distribution board in the middle of the QAV250 frame and fix it. Next, fix the ESC on each arm of the frame, as shown in Fig. 4. Screw all the A2212 motors on the drone frame.

Connect three wires from each ESC to the three wires of each motor. Then connect the three wires of each ESC to CC3D flight controller with +, -, and S



Fig. 1: Author's prototype

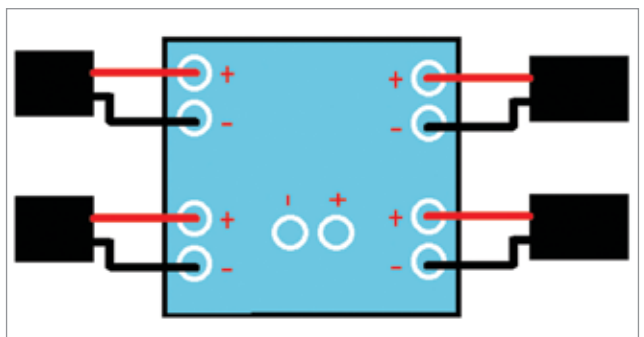


Fig. 3: Power distribution board with XT60 connectors



Fig. 2: Assembled drone frame



Fig. 4: ESCs and motors fixed on arms of the frame